

Predicting Pitcher ERA Using Advanced Metrics

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1. Introduction

```
library(readr)
library(dplyr)

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(corrplot)

## corrplot 0.95 loaded

# Load dataset
Pitchers2025 <- read_csv("Pitchers2025.csv")

## Rows: 69 Columns: 11

## — Column specification

```

```
## Delimiter: ","
## chr (3): Player, Role, Handedness
## dbl (8): K%, BB%, ERA, xwOBA, Avg_FB_Velo, WHIP, GB%, HR9
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this
message.

head(Pitchers2025, 2)

## # A tibble: 2 × 11
##   Player   Role   `K%` `BB%`   ERA Handedness xwOBA Avg_FB_Velo  WHIP `GB%`
HR9
##   <chr>   <chr> <dbl> <dbl> <dbl> <chr>      <dbl>      <dbl> <dbl> <dbl>
<dbl>
## 1 Warren... Star... 24.1   9.1   4.44 Right      0.33        93.3   1.37  42.8
1.2
```

```
## 2 Bello,... Star... 17.7 8.4 3.35 Right 0.327 95.1 1.24 50
0.9
```

This project explored the extent to which underlying pitching performance metrics explain variation in earned run average (ERA) among Major League Baseball pitchers. Rather than relying exclusively on traditional outcome statistics, the analysis focused on process-based metrics that better capture pitcher skill and contact quality.

Using 2025 Statcast data from 69 MLB pitchers, a multiple linear regression model was developed to examine the relationship between ERA and several predictor variables, including expected weighted on-base average (xwOBA), strikeout rate (K%), home run rate (HR/9), and pitcher handedness. Among these variables, xwOBA was expected to be the strongest predictor because it directly reflects the quality of contact allowed by the pitcher.

The purpose of this study was to evaluate whether advanced underlying metrics provide stronger insight into pitching performance than ERA alone while also demonstrating how statistical modeling can be applied within baseball analytics and player evaluation.

2. Model Setup

$$ERA_i = \beta_0 + \beta_1(xwOBA_i) + \beta_2(K\%_i) + \beta_3(HR9_i) + \beta_4(Handedness_i) + \epsilon_i$$

State LINE assumptions. State your depth of understanding regarding the LINE A multiple linear regression framework was used to evaluate the relationship between ERA and several underlying pitching metrics. The response variable in the model was ERA, while the predictor variables included xwOBA, strikeout rate (K%), home run rate (HR/9), and pitcher handedness.

Because handedness represents distinct categorical groups rather than continuous numerical values, it was converted into a factor variable prior to modeling. The regression analysis assumed linear relationships between predictors and ERA, independent observations, normally distributed residuals, and constant residual variance across fitted values.

```
str(Pitchers2025)

## spc_tbl_ [69 × 11] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
## $ Player      : chr [1:69] "Warren, Will" "Bello, Brayan" "Skubal, Tarik"
## $ Role        : chr [1:69] "Starter" "Starter" "Starter" "Starter" ...
## $ K%          : num [1:69] 24.1 17.7 32.2 21.3 29.4 21.7 19.8 23.3 23.6
```

```

22.5 ...
## $ BB%      : num [1:69] 9.1 8.4 4.4 7.1 8.6 6.2 8 8.5 6.4 9.6 ...
## $ ERA      : num [1:69] 4.44 3.35 2.21 4.24 2.49 3.54 4.17 3.66 2.86
3.99 ...
## $ Handedness : chr [1:69] "Right" "Right" "Left" "Right" ...
## $ xwOBA     : num [1:69] 0.33 0.327 0.258 0.297 0.259 0.312 0.337 0.303
0.287 0.327 ...
## $ Avg_FB_Velo: num [1:69] 93.3 95.1 97.6 94.4 95.4 95 93 94.3 95.8 94.8
...
## $ WHIP      : num [1:69] 1.37 1.24 0.89 1.23 0.99 1.18 1.3 1.24 1.1 1.42
...
## $ GB%      : num [1:69] 42.8 50 41.6 45.3 53.7 41.1 42.3 59.4 52.6 40.3
...
## $ HR9      : num [1:69] 1.2 0.9 0.8 1.3 0.7 1.1 1.4 0.7 0.6 1.2 ...
## - attr(*, "spec")=
## .. cols(
## ..   Player = col_character(),
## ..   Role = col_character(),
## ..   `K%` = col_double(),
## ..   `BB%` = col_double(),
## ..   ERA = col_double(),
## ..   Handedness = col_character(),
## ..   xwOBA = col_double(),
## ..   Avg_FB_Velo = col_double(),
## ..   WHIP = col_double(),
## ..   `GB%` = col_double(),
## ..   HR9 = col_double()
## .. )
## - attr(*, "problems")=<externalptr>

```

```

Pitchers2025$Handedness <- factor(Pitchers2025$Handedness,
                                  labels = c("Left", "Right"))

```

```

test_model <- lm(ERA ~ xwOBA + `K%` + HR9 + Handedness, data = Pitchers2025)
summary(test_model)

```

```

##
## Call:
## lm(formula = ERA ~ xwOBA + `K%` + HR9 + Handedness, data = Pitchers2025)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.14239 -0.46278 -0.08996  0.40032  2.00690
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -3.646917   1.850357  -1.971   0.0531 .

```

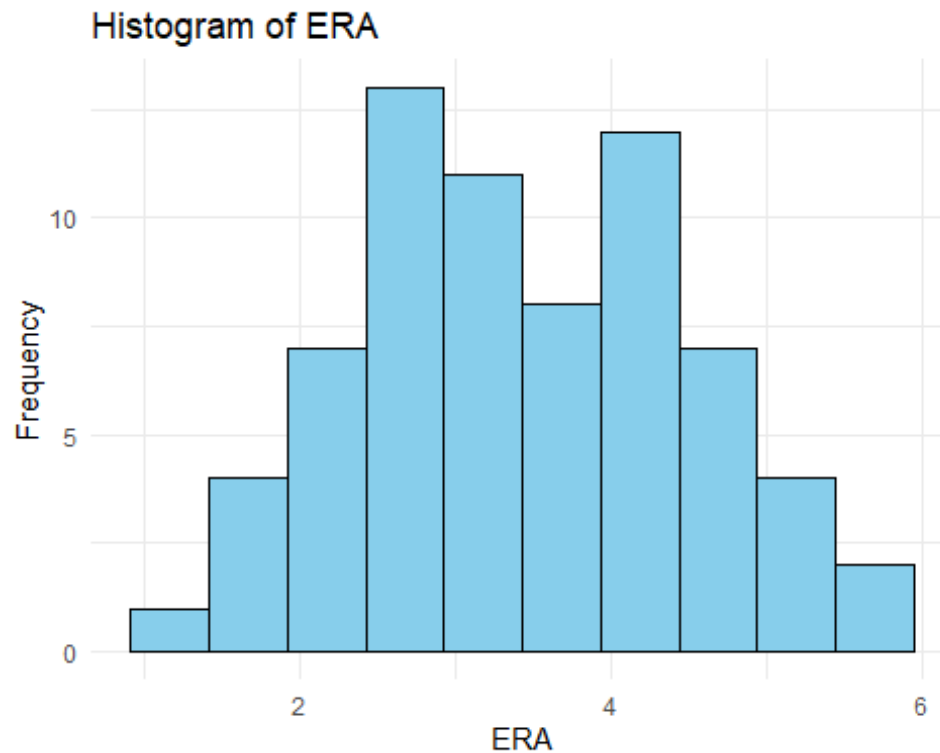
```
## xwOBA          23.529916    5.097672    4.616 1.94e-05 ***
## `K%`          -0.006021    0.021927   -0.275    0.7845
## HR9           0.411031    0.288877    1.423    0.1596
## HandednessRight -0.048932    0.167497   -0.292    0.7711
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6257 on 64 degrees of freedom
## Multiple R-squared:  0.6786, Adjusted R-squared:  0.6585
## F-statistic: 33.78 on 4 and 64 DF,  p-value: 3.812e-15
```

3. Explanatory Data Analysis

Before fitting the regression model, exploratory data analysis was conducted to better understand the distribution and structure of the data. Summary statistics and visualizations were used to evaluate the spread of the variables and identify potential irregularities prior to modeling.

```
library(ggplot2)

ggplot(Pitchers2025, aes(x = ERA)) +
  geom_histogram(bins = 10, fill = "skyblue", color = "black") +
  labs(
    title = "Histogram of ERA",
    x = "ERA",
    y = "Frequency"
  ) +
  theme_minimal()
```



The histogram of ERA shows a roughly symmetric, bell-shaped distribution with a slight right skew.

```
library(dplyr)
library(tidyr)
library(knitr)

output <- Pitches2025 %>%
  summarize(
    across(c(ERA, xwOBA, `K%`, HR9),
      list(Mean = mean, SD = sd, Min = min, Max = max),
      na.rm = TRUE)
  ) %>%
  pivot_longer(
    cols = everything(),
    names_to = c("Variable", "Statistic"),
    names_sep = "_"
  ) %>%
  pivot_wider(
    names_from = Statistic,
    values_from = value
  )

kable(output)
```

Variable	Mean	SD	Min	Max
ERA	3.4289855	1.0708067	1.170	5.70
xwOBA	0.2928116	0.0313178	0.227	0.37
K%	25.8927536	6.3923971	14.200	44.40
HR9	0.9130435	0.3721610	0.200	1.90

```
table(Pitchers2025$Handedness)

##
##  Left Right
##    22    47

output_cat <- prop.table(table(Pitchers2025$Handedness)) %>%
  as.data.frame()

colnames(output_cat) <- c("Handedness", "Proportion")

kable(output_cat)
```

Handedness	Proportion
Left	0.3188406
Right	0.6811594

4. Correlation Analysis & Multicollinearity Problem

```
library(car)

## Loading required package: carData

##
## Attaching package: 'car'

## The following object is masked from 'package:dplyr':
##
##      recode

vif(test_model)

##      xwOBA      `K%`      HR9 Handedness
##  4.426481  3.412140  2.007344  1.073763
```

```
library(dplyr)

df <- Pitchers2025 %>%
```

```

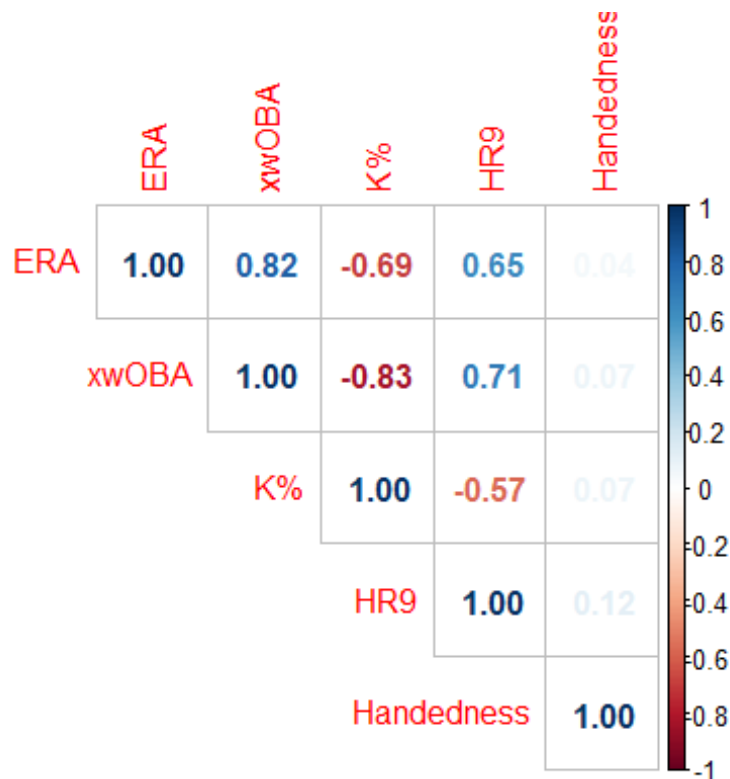
select(ERA, xwOBA, `K%`, HR9)

df$Handedness <- as.integer(Pitchers2025$Handedness)

df_cor <- cor(df)

library(corrplot)
corrplot::corrplot(df_cor, method = "number", type = "upper")

```



Correlation analysis and variance inflation factor (VIF) diagnostics were conducted to evaluate potential multicollinearity among the predictor variables. While xwOBA and strikeout rate demonstrated moderate correlation, all VIF values remained below 5, indicating that multicollinearity was not severe enough to compromise model interpretation.

5. Model Comparison

```

m11 <- lm(ERA ~ xwOBA, data = Pitchers2025)
m12 <- lm(ERA ~ `K%`, data = Pitchers2025)
m13 <- lm(ERA ~ HR9, data = Pitchers2025)
m14 <- lm(ERA ~ Handedness, data = Pitchers2025)

m22 <- lm(ERA ~ xwOBA + `K%`, data = Pitchers2025)
m23 <- lm(ERA ~ xwOBA + HR9, data = Pitchers2025)

```

```

m24 <- lm(ERA ~ xwOBA + Handedness, data = Pitchers2025)

m33 <- lm(ERA ~ xwOBA + `K%` + HR9, data = Pitchers2025)
m34 <- lm(ERA ~ xwOBA + `K%` + Handedness, data = Pitchers2025)

m4 <- lm(ERA ~ xwOBA + `K%` + HR9 + Handedness, data = Pitchers2025)

models <- list(m11, m12, m13, m14, m22, m23, m24, m33, m34, m4)

model_names <- c("m11", "m12", "m13", "m14", "m22", "m23", "m24", "m33", "m34", "m4")

results <- data.frame(
  Model = model_names,
  R2 = sapply(models, function(x) summary(x)$r.squared),
  Adj_R2 = sapply(models, function(x) summary(x)$adj.r.squared)
)

results

```

##	Model	R2	Adj_R2
## 1	m11	0.667819697	0.66286178
## 2	m12	0.477186044	0.46938285
## 3	m13	0.417843254	0.40915435
## 4	m14	0.001894163	-0.01300294
## 5	m22	0.668305383	0.65825403
## 6	m23	0.677559740	0.66778882
## 7	m24	0.668105147	0.65804773
## 8	m33	0.678187230	0.66333433
## 9	m34	0.668449415	0.65314708
## 10	m4	0.678615788	0.65852928

Multiple regression models were compared using both R^2 and adjusted R^2 values to determine which combination of predictors best explained variation in ERA. Although several models produced similar explanatory power, the model containing xwOBA and HR/9 produced the highest adjusted R^2 value (0.6678), suggesting the strongest overall fit while maintaining relative simplicity.

6. Statistical Inferences

```

best_model <- lm(ERA ~ xwOBA + `K%` + HR9 + Handedness, data = Pitchers2025)

library(broom)
library(knitr)

my_output <- tidy(best_model)
kable(my_output)

```

term	estimate	std.error	statistic	p.value
(Intercept)	-3.6469171	1.8503573	-1.9709258	0.0530602
xwOBA	23.5299157	5.0976722	4.6158158	0.0000194
K%	-0.0060205	0.0219272	-0.2745692	0.7845316
HR9	0.4110306	0.2888773	1.4228553	0.1596344
HandednessRight	-0.0489315	0.1674967	-0.2921344	0.7711288

Estimated Regression Model:

$$\widehat{ERA}_i = -3.647 + 23.530(xwOBA_i) - 0.006(K\%_i) + 0.411(HR9_i) - 0.049(RightHanded_i)$$

- $H_0: \beta_j = 0$ (no effect on ERA)
- $H_1: \beta_j \neq 0$ (has an effect on ERA)

Derivation of t-value:

$$t = \frac{\hat{\beta}_j}{SE(\hat{\beta}_j)}$$

- if p-val < 0.05 -> reject H_0
- if p-val $\geq$ 0.05 -> Fail to reject H_0

Statistical Significance:

- xwOBA: Reject H_0 -> Statistically Significant
- K%: Fail to reject H_0 -> NOT Statistically Significant
- HR/9: Fail to reject H_0 -> NOT Statistically Significant
- Handedness: Fail to reject H_0 -> NOT Statistically Significant

xwOBA is statistically significant. A 1-unit increase in xwOBA increases ERA by 23.53; more realistically, a 0.01 increase raises ERA by about 0.235, holding other variables constant.

K% (p = 0.7845), HR/9 (p = 0.1596), and handedness (p = 0.7711) are not statistically significant, so we fail to reject H_0 . This means their effects on ERA are not reliably different from zero. • Whether the variable has no effect or uncertain effect

A large p-value implies insufficient evidence of an effect, not that the variable has no effect, but that its effect is uncertain in this model.

```
kable(confint(best_model))
```

	2.5 %	97.5 %
(Intercept)	-7.3434308	0.0495967
xwOBA	13.3461448	33.7136866
K%	-0.0498251	0.0377841
HR9	-0.1660681	0.9881292
HandednessRight	-0.3835447	0.2856816

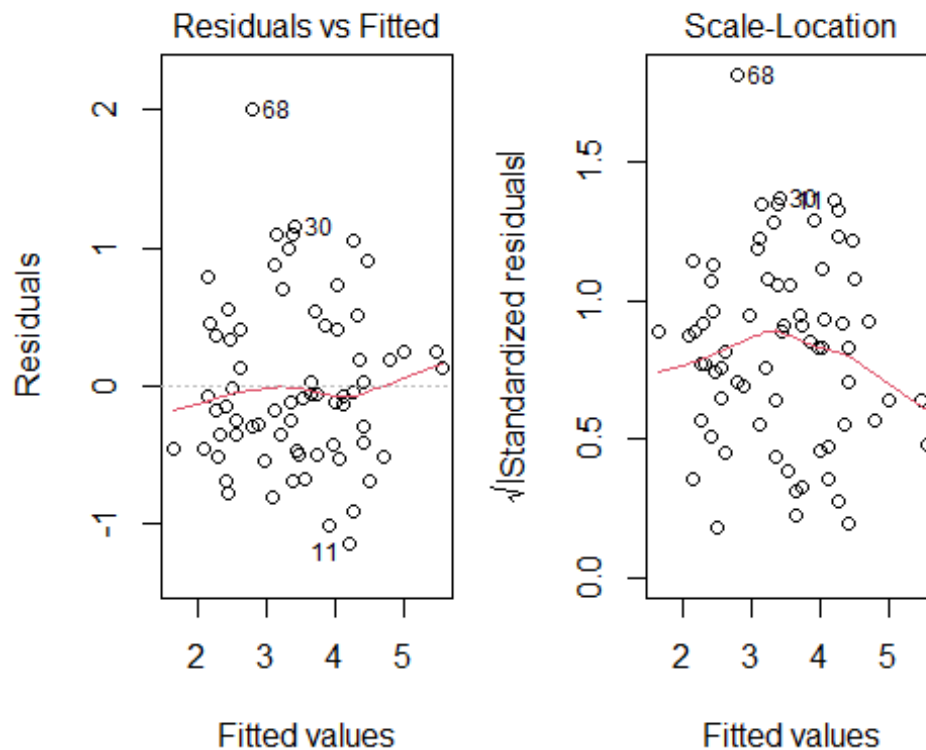
```
new_data <- data.frame(  
  xwOBA = 0.310,  
  `K%` = 22,  
  HR9 = 1.1,  
  Handedness = "Right",  
  check.names = FALSE  
)  
  
predict(best_model, new_data, interval = "confidence")  
  
##           fit          lwr          upr  
## 1 3.918107 3.709044 4.12717
```

We are 95% confident that the true mean ERA for pitchers with xwOBA = 0.310, K% = 22, HR/9 = 1.1, and right-handedness lies between 3.709 and 4.127.

7. Diagnostics

Residual vs Fitted + Scale Location

```
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))  
  
plot(best_model, which = 1) # Residuals vs Fitted  
plot(best_model, which = 3) # Scale-Location
```



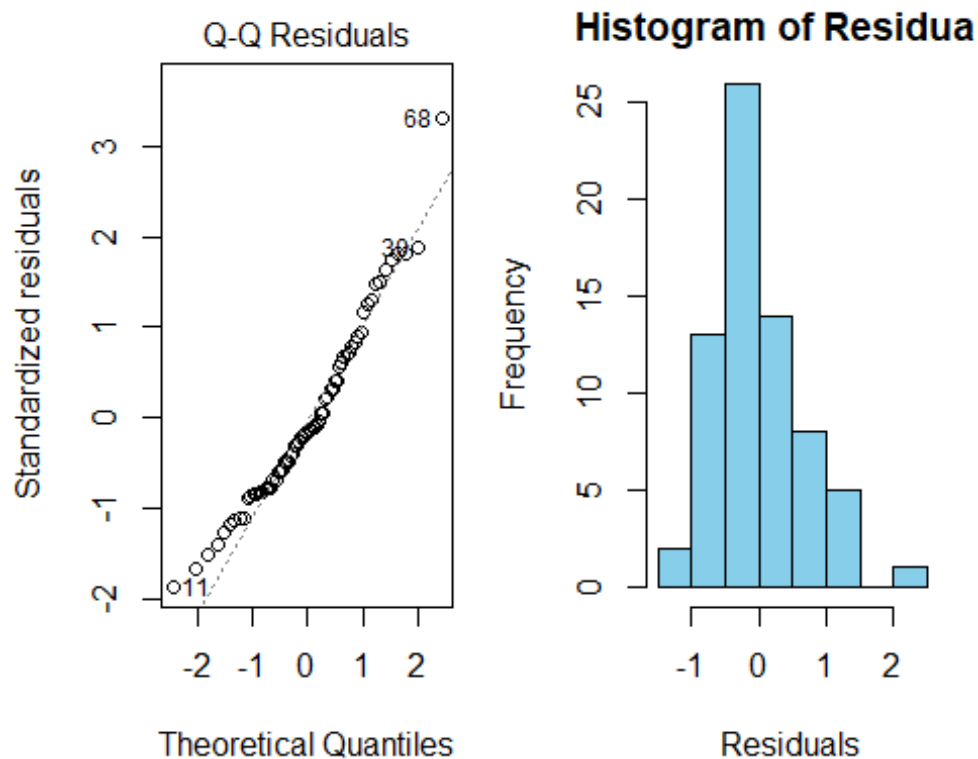
The Residuals vs Fitted and Scale-Location plots were used to evaluate the assumptions of linearity and constant variance (homoscedasticity). The residuals appeared to be mostly randomly scattered around zero, although a slight curved pattern was present, suggesting mild evidence of non-linearity. In addition, the Scale-Location plot showed some variation in the spread of residuals across fitted values, indicating mild heteroscedasticity. Overall, the assumptions partially held, though there was slight evidence that the model could be improved. One potential improvement would be applying a transformation, such as a logarithmic transformation, to stabilize variance and strengthen the linear relationship between the predictors and ERA.

Q-Q Plot + Histo of Residuals

```
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))

plot(best_model, which = 2) # Normal Q-Q Plot

hist(best_model$residuals,
      breaks = 10,
      main = "Histogram of Residuals",
      xlab = "Residuals",
      col = "skyblue",
      border = "black")
```

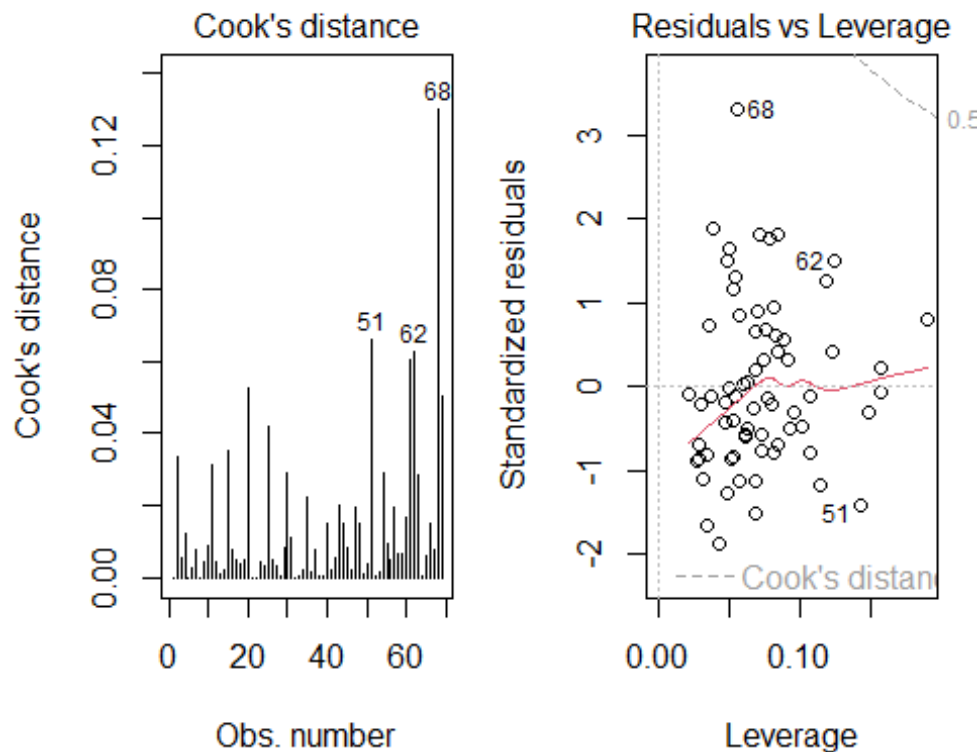


The Normal Q-Q plot and histogram of residuals were used to assess the normality assumption of the regression model. In the Q-Q plot, most points closely followed the reference line, though some deviations were observed at the tails, suggesting slight departures from normality. Similarly, the histogram of residuals appeared approximately bell-shaped with mild skewness. Taken together, these results indicate that the normality assumption partially held. A potential improvement to the model would be to investigate or remove potential outliers, as influential residual values can distort the distribution of residuals and impact statistical inference.

Cook's Distance + Residuals vs Leverage

```
par(mfrow = c(1, 2), mar = c(4, 4, 2, 1))

plot(best_model, which = 4) # Cook's Distance
plot(best_model, which = 5) # Residuals vs Leverage
```



Cook's Distance and Residuals vs Leverage plots were used to identify influential observations within the regression model. Several observations, particularly points 68, 51, and 62, displayed relatively higher leverage and Cook's Distance values compared to the rest of the dataset. Although these observations were somewhat influential, none appeared extreme enough to suggest severe model instability. Therefore, the assumption regarding influential observations generally held, though some caution is warranted. One possible improvement would be to analyze the model both with and without these observations to evaluate how strongly they influence the regression estimates and overall model performance.

8. Conclusion

Overall, the regression analysis revealed that ERA was most strongly associated with xwOBA, while strikeout rate (K%), home run rate (HR/9), and pitcher handedness demonstrated weaker relationships with the response variable. Among all predictors included in the model, xwOBA emerged as the most influential variable and was the only predictor found to be statistically significant. In contrast, K%, HR/9, and handedness were not statistically significant, suggesting that their effects on ERA were not reliably different from zero within this particular model and dataset.

From a practical baseball perspective, these findings reinforce the importance of contact quality in pitcher performance evaluation. Pitchers who allowed higher-quality contact, as

measured by xwOBA, generally produced higher ERA values, indicating poorer run prevention outcomes. These results suggest that advanced expected statistics may provide more meaningful insight into pitching effectiveness than traditional outcome-based metrics alone. In practice, the model could be used to support pitcher evaluation and player development efforts by helping identify pitchers who consistently limit quality contact.

Despite the model producing strong explanatory power, several limitations should be acknowledged. ERA is influenced by external factors such as defensive performance, sequencing, and luck, which are not fully captured within the regression framework. Additionally, several influential observations were identified during diagnostic analysis, which may have impacted the regression estimates. Future improvements could include incorporating metrics such as FIP or xERA, expanding the sample size, or addressing influential observations to further improve model accuracy and interpretability.